Chapter 10

OPERATIONS RESEARCH

Learning Objectives

After studying this chapter, the students will be able to understand

- Formulation of Linear Programming Problem
- Solution of LPP by graphical method.
- Construction of network of a project
- Project completion time by Critical Path Method(CPM)

Introduction


L.V. Kantorovich

During the world war II, the military management in England recruited a team of scientists, engineers and mathematicians to study the strategic and tactical problems of air and land defence. Their objective was to determine the best utilization of limited military resources like ammunition, food and other things needed for war. This group of scientists formed the first operations research team. The name operations research was apparently coined because the team was dealing with the research on (military) operations. Operations research team helped in developing strategies for mining operations, inventing new flight pattern and planning of sea mines. Following the end of war the success of military team attracted the attention of industrial managers who were seeking solutions to their complex type of problems.

It is not possible to give uniformly acceptable definition of operations research. The following is the definition of operations research published on behalf of UK operational research society. "operations research is the application of scientific methods to complex problems arising from operations involving large system of men, machines, materials and money in industry, business, government and defence."

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An operations research model is defined as any abstract or idealized representation of real life system or situation. The objective of the model is to identify significant factors and inter-relationship. Here we study only two models namely linear programming problem and network analysis.

10.1 Linear Programming Problem

The Russian Mathematician L.V. Kantorovich applied mathematical model to solve linear programming problems. He pointed out in 1939 that many classes of problems which arise in production can be defined mathematically and therefore can be solved numerically. This decision making technique was further developed by George B. Dantziz. He formulated the general linear programming problem

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